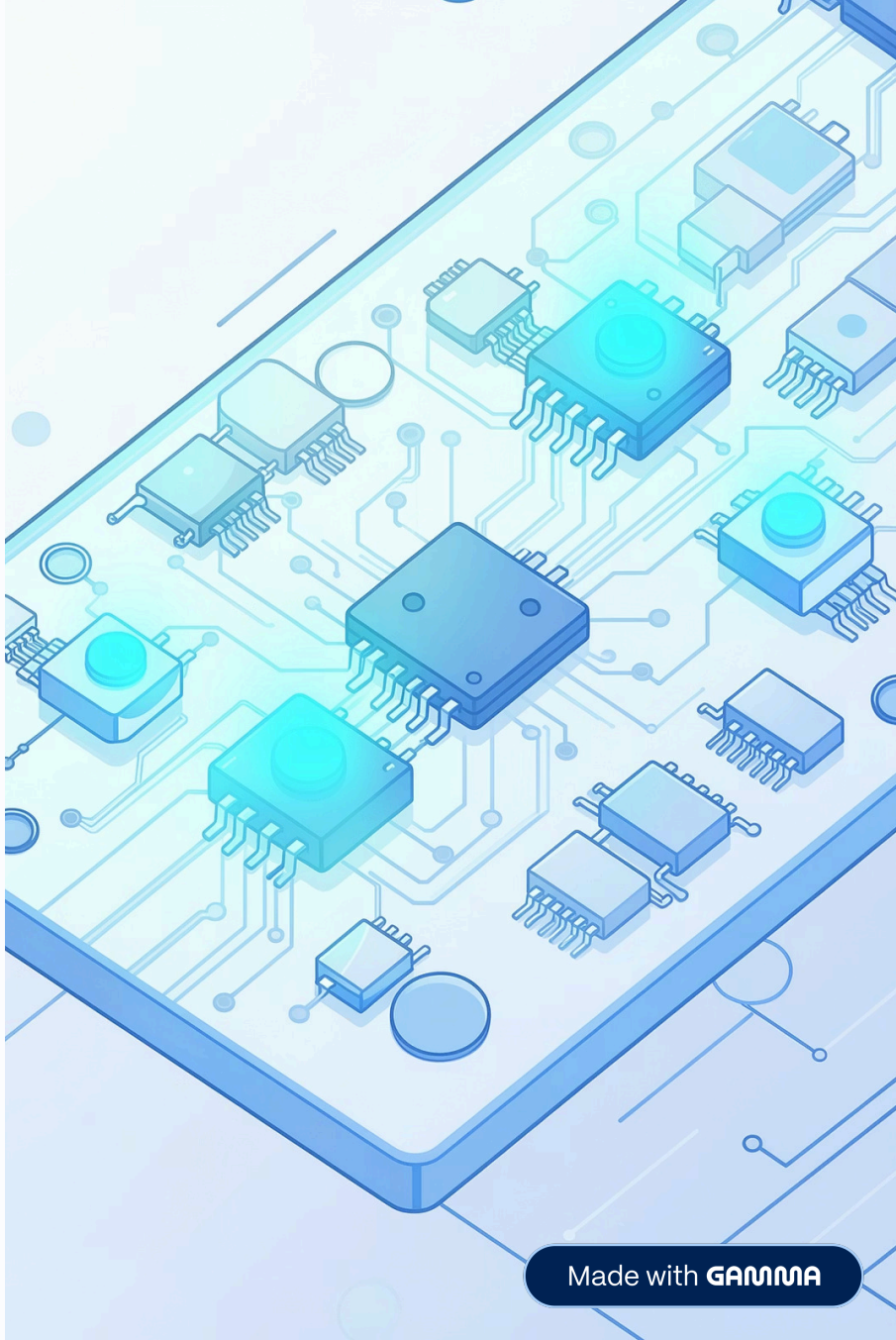
Bipolar Junction Transistors (BJTs)

An essential guide to one of electronics' most foundational components — from structure and operation to real-world applications.

SEMICONDUCTOR FUNDAMENTALS



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What Is a BJT?

A **Bipolar Junction Transistor (BJT)** is a three-terminal semiconductor device that uses both electrons and holes as charge carriers. It can amplify current or act as a fast electronic switch — making it indispensable in analog and digital circuits alike.

Three Terminals

Emitter, Base, and Collector

Current-Controlled

Base current controls collector current

Bipolar

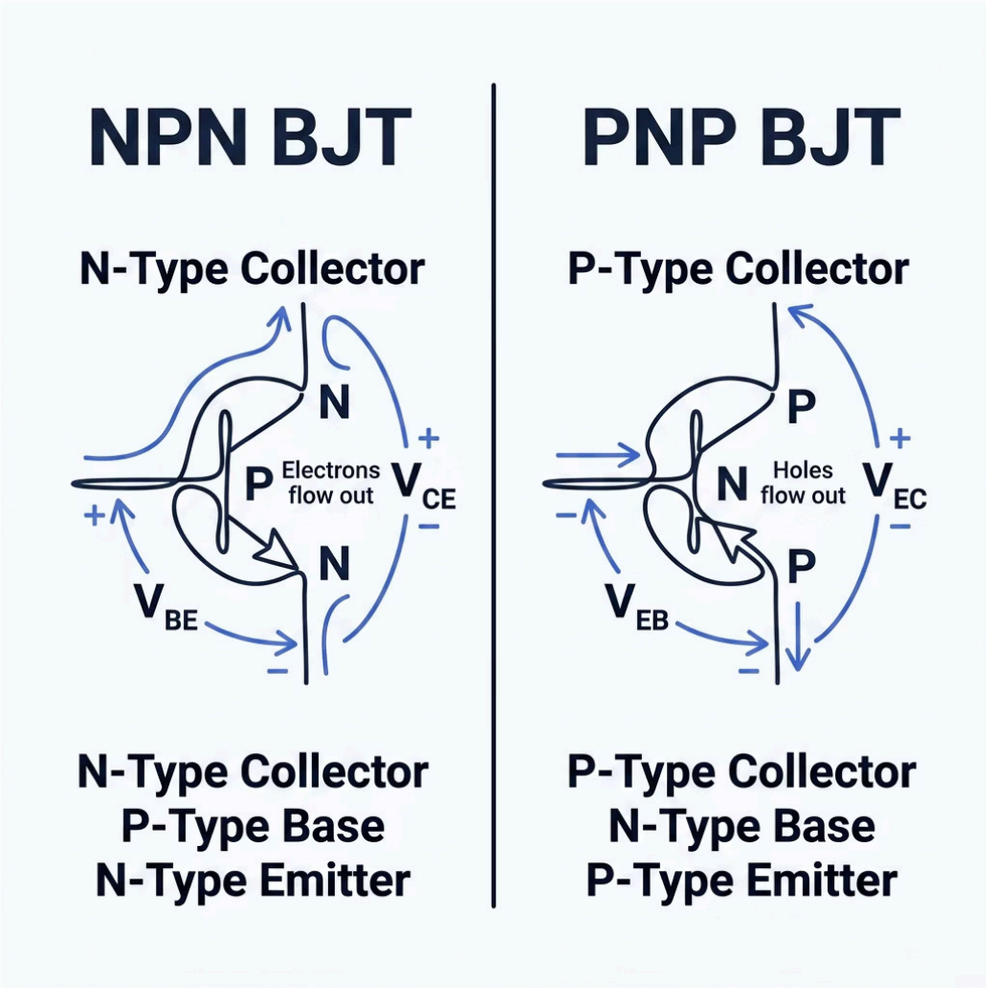
Uses both electrons and holes

BJT Structure & Symbol

NPN Configuration

A thin **P-type base** sandwiched between two N-type regions. Current flows from collector to emitter when the base is forward-biased. Electrons are the majority carriers.

MOST COMMONLY USED



PNP is the complementary type — current flows from emitter to collector, and the base must be pulled *low* to turn it on.

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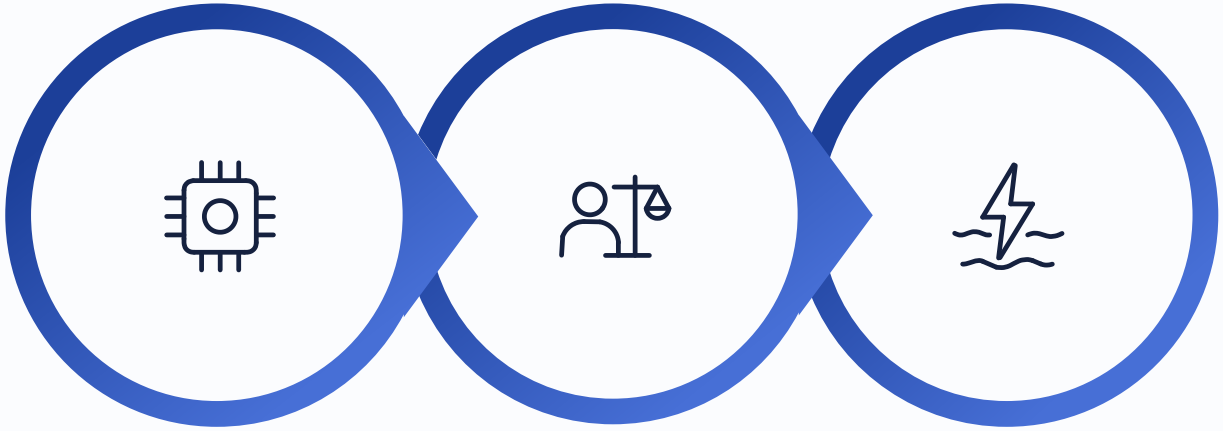
How a BJT Works

A small current injected into the **base** controls a much larger current between collector and emitter — this is **current amplification**.



Small Base Current

Emitter Forward-Bias

Large Collector Current

The ratio of collector current to base current is called **β (beta)** or **hFE**, typically ranging from 20 to over 500 depending on the transistor.

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BJT Operating Regions

1

Cutoff

Base-emitter junction not forward-biased. **No current flows** — the transistor is fully *off*, acting as an open switch.

2

Active

Base-emitter forward-biased, base-collector reverse-biased. **Ic = β × Ib** — the transistor amplifies linearly.

3

Saturation

Both junctions forward-biased. **VCE is minimal** — the transistor is fully *on*, acting as a closed switch.

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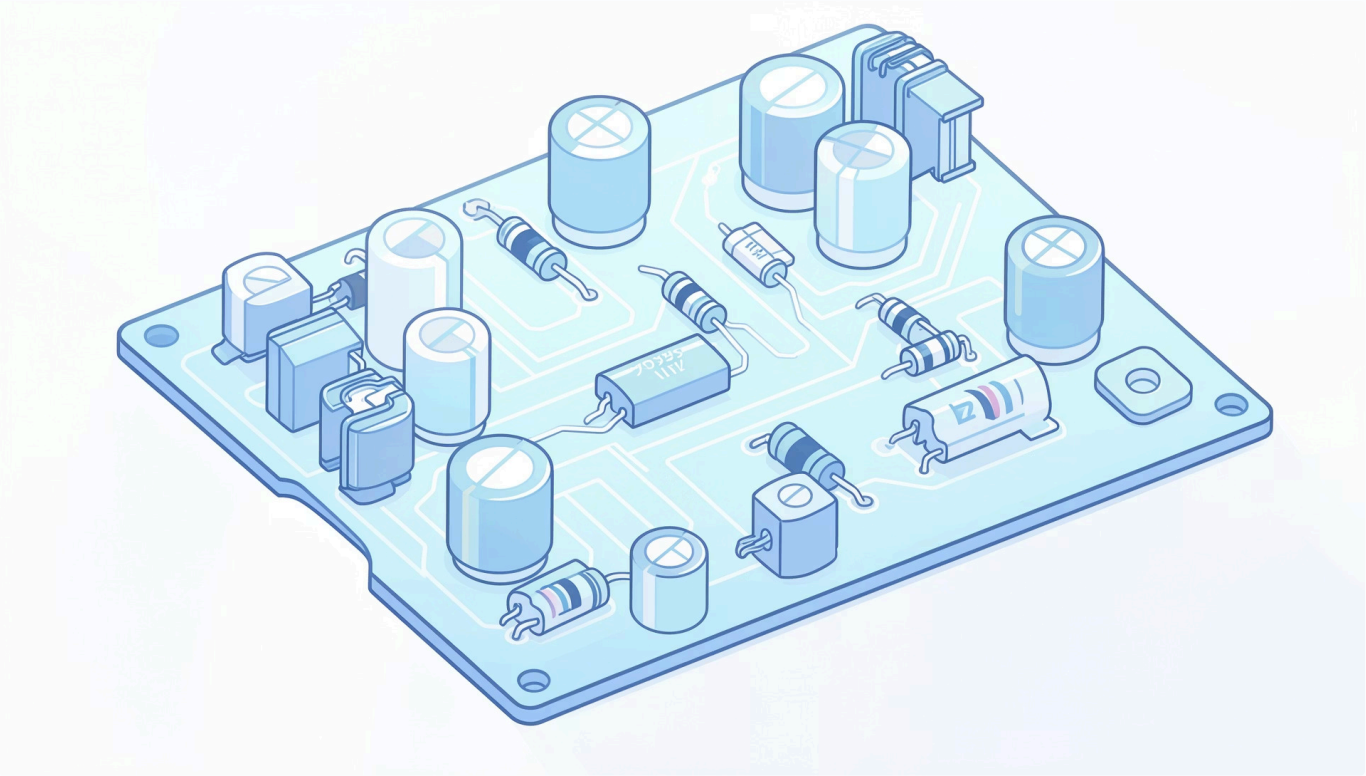

ANALOG MODE

BJT as an Amplifier

In the **active region**, a BJT amplifies small AC signals. Proper **DC biasing** sets the Q-point (quiescent point) to keep the transistor in its linear range.

Small-Signal Gain

Voltage gain $A_v \approx -g_m \times R_C$, where g_m is transconductance

Biasing Networks

Voltage-divider bias stabilizes the Q-point against temperature and β variations

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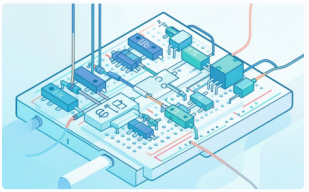
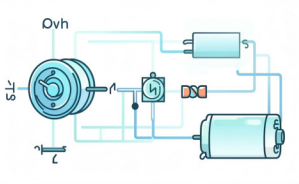
BJT as a Switch

In digital and power circuits, BJTs toggle between **cutoff** (OFF) and **saturation** (ON) — enabling fast, efficient switching with no linear amplification.



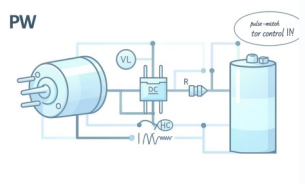
Digital Logic

BJTs form the basis of TTL logic gates, switching in nanoseconds for high-speed computing.



Power Switching

Drive motors, LEDs, and relays by switching high currents with a small control signal.



PWM Control

Pulse-width modulation uses rapid BJT switching to control average power delivery efficiently.

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Key BJT Parameters

β (hFE) — Current Gain

Ratio of Ic to Ib. Varies with temperature, current level, and between individual transistors.

IC(max) — Collector Current

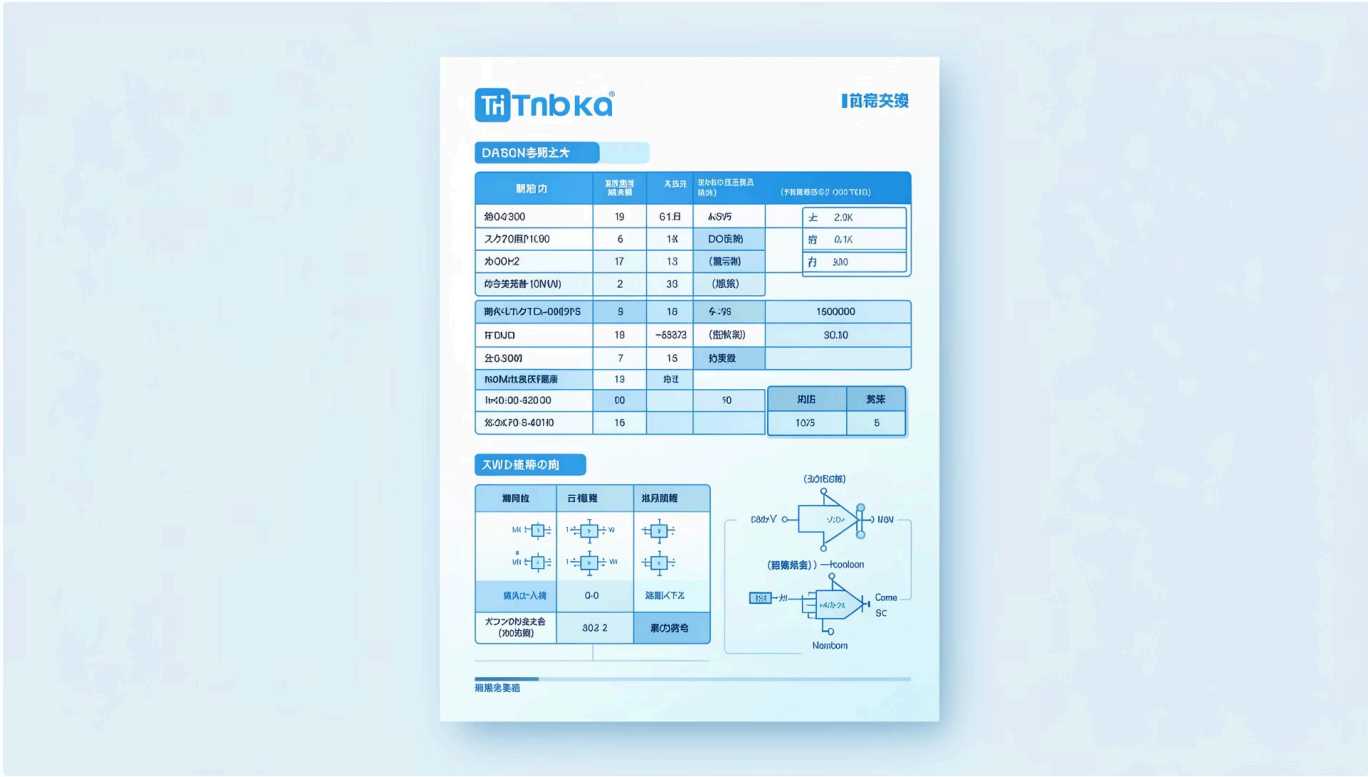
Maximum continuous collector current before thermal damage occurs.

VCEO — Breakdown Voltage

Maximum collector-emitter voltage with base open. Exceeding it causes avalanche breakdown.

fT — Transition Frequency

Frequency at which current gain drops to unity — defines high-frequency limits.



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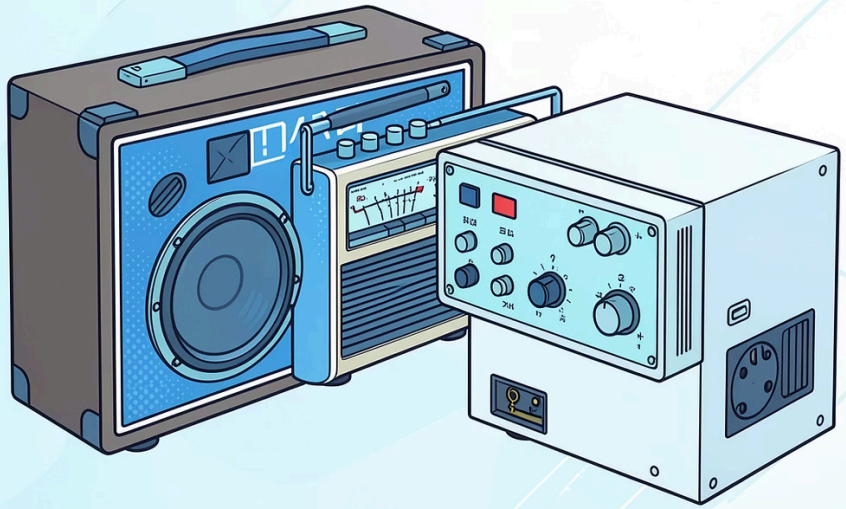

Common BJT Applications

Audio Amplifiers

Class A, B, and AB amplifier stages in hi-fi systems and public address equipment.

Linear Regulators

Pass transistors in linear voltage regulators provide stable, low-noise DC output.

RF & Oscillators

High-frequency BJTs used in radio transmitters, receivers, and oscillator circuits.

Sensor Interfaces

Signal conditioning and amplification for temperature, light, and pressure sensors.

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The Enduring Relevance of BJTs

> Despite the rise of MOSFETs and integrated circuits, the BJT remains a cornerstone of analog electronics — valued for its linearity, speed, and simplicity.

🔬 Educational Foundation

Essential for understanding semiconductor physics and circuit theory

⚡ Analog Performance

Superior linearity and transconductance in precision analog designs

🏭 Still in Production

Billions of BJTs manufactured yearly for industrial, automotive, and consumer electronics

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